

\$ Linux Terminal Basics

A beginner's cheat sheet for the Linux command line

■ Linux is case-sensitive!

01 Understanding the Terminal Prompt

```
usman369@LinuxMint:~$ ← you type your commands here
```

```
usman369 → your username
```

```
LinuxMint → your device / hostname
```

```
~ → active user's home directory
```

```
$ → regular user (root uses #)
```

Terminal shortcut: **Ctrl + Alt + T**

02 Getting Help

■ **man** (Manual)

```
usman369@LinuxMint:~$ man <command>
```

Opens the full user manual for any command. Press Q to quit.

■ **help**

```
usman369@LinuxMint:~$ <command> --help
```

Displays a quick summary of syntax, usage, and available options.

■ **clear**

```
usman369@LinuxMint:~$ clear
```

Clears everything currently visible in the terminal.

03 Navigation

■ **pwd** (Print Working Directory)

```
usman369@LinuxMint:~$ pwd
```

Shows the full path of your current directory, from root to where you are now.

■ **ls** (List)

```
usman369@LinuxMint:~$ ls
```

Lists all visible files and subdirectories in the current directory.

```
usman369@LinuxMint:~$ ls -a OR la
```

Lists everything, including hidden files (those starting with a dot).

```
usman369@LinuxMint:~$ ls <letter/word>* OR ls *<letter/word/.extension>
```

Filters the list by a starting or ending letter, word, or file extension using wildcards.

```
usman369@LinuxMint:~$ ls -l
```

Shows files with their creation / modification timestamps.

```
usman369@LinuxMint:~$ ls -lu
```

Shows files with their last access timestamps.

■ **cd** (Change Directory)

```
usman369@LinuxMint:~$ cd <directory>/
```

Moves into a subdirectory of the current location. e.g. cd Downloads/

```
usman369@LinuxMint:~$ cd ..
```

Goes one directory level back (up to the parent).

```
usman369@LinuxMint:~$ cd -
```

Goes back to the previous directory. Use it again to toggle between the two.

```
usman369@LinuxMint:~$ cd ~ OR cd
```

Takes you directly to the active user's home directory.

```
usman369@LinuxMint:~$ cd /
```

Takes you to the root directory (equivalent to C:\ in Windows).

```
usman369@LinuxMint:~$ cd /<directory1>/<directory2>/...
```

Navigates to any directory using its absolute path from root.

■ *Press Tab key once to autocomplete a name. Twice to list all matching options.*

04 Creating Files & Directories

■ **mkdir** (Make Directory)

```
usman369@LinuxMint:~$ mkdir <directory1> <directory2> ...
```

Creates one or more directories in the current location.

```
usman369@LinuxMint:~$ mkdir <directory>{1..5}
```

Creates multiple directories in a numbered range, e.g. folder1 through folder5.

```
usman369@LinuxMint:~$ mkdir <directory> -v
```

Verbose mode — confirms whether the directory was created or already exists.

```
usman369@LinuxMint:~$ mkdir -p <directory1>/<directory2>/...
```

Creates a full nested directory path in one command.

■ **touch**

```
usman369@LinuxMint:~$ touch <file1> <file2> ...
```

Creates one or more empty files. Cannot be used to edit file contents.

```
usman369@LinuxMint:~$ touch <file>{1..5}
```

Creates multiple files in a numbered range.

```
usman369@LinuxMint:~$ touch -a <file>
```

Updates the access timestamp of the file to the current time.

```
usman369@LinuxMint:~$ touch -m <file>
```

Updates the modification timestamp of the file to the current time.

```
usman369@LinuxMint:~$ touch -r <file1> <file2>
```

Copies the timestamps from file1 and applies them to file2.

05 Viewing & Editing Files

■ **cat** (Concatenate)

```
usman369@LinuxMint:~$ cat > <file>
```

Creates a new file and lets you type content into it. Press Ctrl+D when done.

```
usman369@LinuxMint:~$ cat >> <file>
```

Appends new text to the end of an existing file. Press Ctrl+D when done.

```
usman369@LinuxMint:~$ cat <file1> <file2> ...
```

Displays the contents of one or more files in the terminal.

```
usman369@LinuxMint:~$ cat <file1> > <file2>
```

Copies the contents of file1 into file2. Creates file2 if it doesn't exist.

```
usman369@LinuxMint:~$ cat -n <file>
```

Displays file contents with line numbers.

```
usman369@LinuxMint:~$ cat -e <file>
```

Marks the end of each line with a dollar sign (\$).

```
usman369@LinuxMint:~$ cat -v <file>
```

Displays non-printable characters as visible symbols — useful for debugging.

■ **less**

```
usman369@LinuxMint:~$ less <file>
```

Opens files in a scrollable viewer. Press Q to quit.

■ **more**

```
usman369@LinuxMint:~$ more <file>
```

Similar to less — displays large files page by page.

■ **nano** (Default text editor)

```
usman369@LinuxMint:~$ nano <file>
```

Opens a file in the nano text editor inside the terminal. Creates the file if it doesn't exist.

■ **vi** (Text editor)

```
usman369@LinuxMint:~$ vi <file>
```

Opens a file in the vi text editor inside the terminal. Creates the file if it doesn't exist.

■ *Good luck finding how to quit vi.*

■ **head**

```
usman369@LinuxMint:~$ head -<number of lines> <file>
```

Displays the first n lines of the file.

■ **tail**

```
usman369@LinuxMint:~$ tail -<number of lines> <file>
```

Displays the last n lines of the file.

06 Copying, Moving & Deleting

■ **cp** (Copy)

```
usman369@LinuxMint:~$ cp <file/directory> <directory>/
```

Copies a file or directory to another location. The original remains intact. Creates the destination if it doesn't exist.

```
usman369@LinuxMint:~$ cp ../<file/directory> .
```

Copies a file from the parent directory into the current one. The dot (.) at the end means current directory.

■ **mv** (Move)

```
usman369@LinuxMint:~$ mv <file/directory> <directory>/
```

Moves a file or directory to another location. The original is removed. Also used to rename files.

■ **rm / rmdir** (Remove)

```
usman369@LinuxMint:~$ rm <file1> <file2> ...
```

Deletes one or more files.

```
usman369@LinuxMint:~$ rm <file>{1..5}
```

Deletes multiple files in a numbered range.

```
usman369@LinuxMint:~$ rmdir <directory1> <directory2> ...
```

Deletes one or more empty directories.

```
usman369@LinuxMint:~$ rm -r <directory>
```

Recursively deletes a directory and all of its contents.

```
usman369@LinuxMint:~$ rm -f <file/directory>
```

Force-deletes a file or directory and its contents, even if they are in use.

■ *You can combine flags: **rm -rf** — use with extreme caution!*

07 Text Processing

■ **sort**

```
usman369@LinuxMint:~$ sort <file>
```

Sorts the lines of a file alphabetically from A to Z.

```
usman369@LinuxMint:~$ sort -r <file>
```

Sorts the lines in reverse alphabetical order, Z to A.

■ **uniq (Unique)**

```
usman369@LinuxMint:~$ sort <file> | uniq
```

Displays only unique lines from a file. Always sort first — `uniq` only removes consecutive duplicates.

■ **grep (Global Regular Expression Print)**

```
usman369@LinuxMint:~$ grep "<content>" <file1> <file2> ...
```

Searches for and displays every line in the file(s) that contains the given content.

```
usman369@LinuxMint:~$ grep -E "<content1>|<content2>..." <file1> <file2> ...
```

Searches for multiple contents at once across one or more files.

■ **wc (Word Count)**

```
usman369@LinuxMint:~$ wc -l <file>
```

Counts the number of lines in the file.

```
usman369@LinuxMint:~$ wc -w <file>
```

Counts the number of words in the file.

```
usman369@LinuxMint:~$ wc -m <file>
```

Counts the number of characters in the file.

■ **split**

```
usman369@LinuxMint:~$ split -l <number of lines> <file>
```

Splits the file into smaller files of `n` lines each. The original file is not affected.

■ **shuf (Shuffle)**

```
usman369@LinuxMint:~$ shuf <file>
```

Randomly shuffles and displays all lines of the file.

```
usman369@LinuxMint:~$ shuf -n <number of lines> <file>
```

Randomly picks and displays `n` lines from the file.

■ **cmp (Compare)**

```
usman369@LinuxMint:~$ cmp <file1> <file2>
```

Compares two files byte by byte and reports if there is any difference.

■ **diff (Difference)**

```
usman369@LinuxMint:~$ diff <file1> <file2>
```

Compares two files and displays the lines that differ between them.

08 Finding Files

■ **find**

```
usman369@LinuxMint:~$ find <source-path>/ -name <file/directory>
```

Searches for a file or directory by name starting from the given path. Supports wildcards.

■ **locate**

```
usman369@LinuxMint:~$ locate <file/directory>
```

Quickly finds a file or directory by name using a pre-built database. Run updatedb first.

■ **updatedb**

```
usman369@LinuxMint:~$ updatedb
```

Rebuilds the file name database used by the locate command. Run this before using locate.

09 Compression & Archives

■ gzip

```
usman369@LinuxMint:~$ gzip <file>
```

Compresses the file and saves it with a .gz extension. The original file is replaced.

```
usman369@LinuxMint:~$ gzip -k <file>
```

Compresses the file but keeps a copy of the original.

```
usman369@LinuxMint:~$ gzip -d <file.gz>
```

Decompresses a .gz file and restores it to its original form.

■ gunzip

```
usman369@LinuxMint:~$ gunzip <file.gz>
```

Decompresses a .gz file. Equivalent to gzip -d.

■ tar (Tarball — archive files)

```
usman369@LinuxMint:~$ tar -cvf <archive-name.tar> <file1/directory1> ...
```

Creates a new .tar archive from the given files or directories. (-c = create, -v = verbose, -f = output file name)

```
usman369@LinuxMint:~$ tar -xvf <archive-name.tar>
```

Extracts the contents of a .tar archive into the current directory. (-x = extract, -v = verbose, -f = file to read)

```
usman369@LinuxMint:~$ tar -xvf <archive-name.tar> -C <destination-path>/
```

Extracts the archive into a specific destination directory instead of the current one.

```
usman369@LinuxMint:~$ tar -cvzf <archive-name.tar.gz> <file1/directory1> ...
```

Creates a compressed .tar.gz archive (combines tar and gzip in one step). (-z = use gzip compression)

```
usman369@LinuxMint:~$ tar -xvzf <archive-name.tar.gz>
```

Extracts a compressed .tar.gz archive.

```
usman369@LinuxMint:~$ tar -tvf <archive-name.tar>
```

Lists the contents of an archive without extracting anything. (-t = list / table of contents)

■ Think of .tar as a folder that packs everything together, and -z as adding a zip on top of it.

10 Utilities & Miscellaneous

■ **echo**

```
usman369@LinuxMint:~$ echo "<string>"
```

Prints the given string directly to the terminal. Useful in scripts.

■ **date**

```
usman369@LinuxMint:~$ date
```

Displays the current day, month, date, time, time zone, and year.

■ **cal** (Calendar)

```
usman369@LinuxMint:~$ cal
```

Shows the calendar for the current month.

```
usman369@LinuxMint:~$ cal <year>
```

Shows the full calendar for the specified year.

```
usman369@LinuxMint:~$ cal <month> <year>
```

Shows the calendar for a specific month and year.

■ **bc** (Basic Calculator)

```
usman369@LinuxMint:~$ bc
```

Opens a text-based calculator in the terminal. Press Ctrl+D to quit.

■ **history**

```
usman369@LinuxMint:~$ history
```

Displays a numbered list of all previously used commands in the current session.

■ **script**

```
usman369@LinuxMint:~$ script
```

Records everything typed in the terminal until you press Ctrl+D. Output is saved to a file called 'typescript' in the current directory.

■ **alias**

```
usman369@LinuxMint:~$ alias <shortcut>="<command>"
```

Creates a custom shortcut for a long or frequently used command. The alias is removed when you log out.

■ **which**

```
usman369@LinuxMint:~$ which <command>
```

Displays the full executable path of the given command.

■ **uptime**

```
usman369@LinuxMint:~$ uptime
```

Shows how long the system has been running, along with load average information.

■ **neofetch**

```
usman369@LinuxMint:~$ neofetch
```

Displays a stylish summary of your system info (OS, kernel, CPU, RAM, etc.) alongside your distro logo.

11 Process Management

■ `top / ps ax`

```
usman369@LinuxMint:~$ top
```

Shows a live, updating list of all running processes — like Task Manager in Windows.

```
usman369@LinuxMint:~$ ps ax
```

Shows a static snapshot of all running processes at the time of the command.

■ `kill` (Process Ending Command)

```
usman369@LinuxMint:~$ kill <process ID>
```

Terminates a background process by its process ID. Find the ID using `top` or `ps ax`.

■ *Always double-check the process ID before running kill. There is no undo!*

12 Linux File System

Linux uses file systems like **ext3**, **ext4**, **Btrfs**, **XFS**, **ZFS**, etc to store and organise data. Linux Mint uses **ext4**. In Linux, **everything is treated as a file** — including hardware devices and system processes.

Key directories at a glance:

<code>/</code>	Root of the entire file system (like C:\ in Windows)
<code>/home</code>	Contains personal directories for each user
<code>/root</code>	Home directory of the root (admin) user
<code>/etc</code>	System-wide configuration files
<code>/bin</code>	Essential command binaries (ls, cp, mv, etc.)
<code>/usr</code>	User-installed programs and utilities
<code>/var</code>	Variable data — logs, caches, spool files
<code>/tmp</code>	Temporary files, cleared on reboot
<code>/dev</code>	Device files (hard drives, USB, etc.)
<code>/proc</code>	Virtual filesystem showing live process / kernel info
<code>/mnt</code>	Mount point for external drives and partitions

◆ Notes & Credits ◆

This cheat sheet was put together from personal notes taken by me (Usman Ghani) while learning Linux.

A big thanks to the creators whose tutorials made this possible:

MPrashant → youtube.com/@MPrashant

NetworkChuck → youtube.com/@NetworkChuck

"In Linux, everything is a file."